

Advanced AI: Environment Setup

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1 Purpose

This guide installs the **Advanced AI** teaching environment: Python, PyTorch, Jupyter, and supporting libraries so you can run the lab notebooks on Windows, Linux, or macOS. The repository contains `labs/` (notebooks), `notes/` (PDF modules), `datasets/` (sample data), `runtime_env.py`, `configure_kernel.py`, `download_datasets.py`, `requirements.txt`, and platform setup scripts (`setup.bat` / `setup.sh`, `env.ps1` / `env.sh`).

2 What You Need

2.1 Hardware

- Windows 10/11, Linux, or macOS (64-bit).
- NVIDIA GPU with a recent driver (recommended). A 4 GB card works if you use the batch sizes given in each module. CPU-only execution is supported but slower.
- 16 GB RAM minimum (32 GB if you run Spark and Jupyter together).
- About 15 GB free disk space for the virtual environment, PyTorch, and cached model weights.

2.2 Install Before You Start

1. **Python 3.10 or 3.11** from <https://www.python.org/downloads/>. Enable “Add Python to PATH” during installation.
2. **NVIDIA driver** supporting CUDA 12.x (driver version 525 or newer is a safe baseline for CUDA 12.1 wheels).
3. **Microsoft OpenJDK 17** (only required for the Spark lab in Module 18): <https://learn.microsoft.com/en-us/java/openjdk/download>.

3 First-Time Installation

Open a terminal in the course folder (the directory that contains `setup.bat` or `setup.sh`).

3.1 Option A: automated setup

```
setup.bat          # Windows
./setup.sh         # Linux / macOS
```

This creates a virtual environment, installs PyTorch and `requirements.txt`, registers the Jupyter kernel via `configure_kernel.py`, and prefetches bundled datasets with `download_datasets.py --minimal`. Wait until it finishes without errors.

3.2 Option B: manual install

If the setup script fails, run these commands one block at a time:

```
python -m venv .env
.env/Scripts/activate      # Windows
source .env/bin/activate   # Linux / macOS
python -m pip install --upgrade pip setuptools wheel
pip install torch torchvision torchaudio ^
--index-url https://download.pytorch.org/whl/cu121
pip install -r requirements.txt
python configure_kernel.py
python download_datasets.py --minimal
```

For Module 18 (Spark), install JDK 17 and ensure `tools/hadoop/bin/winutils.exe` exists on Windows. After installing JDK, set `JAVA_HOME` to the JDK path (see Section 4).

4 Every Session

Each time you open a new terminal, run one command from the course folder:

```
./env.ps1      # Windows (PowerShell)
./env.sh       # Linux / macOS
```

This activates the virtual environment, sets Hugging Face / Spark / MLflow paths, starts the MLflow UI in the background, and opens Jupyter Lab in your browser. Use kernel **Python 3.11 (labs)**. Press Ctrl+C in the terminal to stop Jupyter; MLflow keeps running in its own window.

To load environment variables without starting servers:

```
./env.ps1 -EnvOnly      # Windows
source ./env.sh --env-only # Linux / macOS
```

Confirm the GPU is visible:

```
python -c "import torch; print(torch.cuda.is_available())"
python -c "import torch; print(torch.cuda.get_device_name(0) if torch.cuda.is_available() else
↪ 'CPU only')"
```

If this prints `False`, PyTorch was installed without CUDA. Reinstall:

```
pip install torch torchvision torchaudio ~
--index-url https://download.pytorch.org/whl/cu121
```

5 Running Labs

If you used `./env.ps1`, Jupyter is already running. Otherwise start it manually:

```
./env.ps1
```

Select kernel **Python 3.11 (labs)** (or the kernel registered during setup). Open notebooks from `labs/`. Read the matching PDF in `notes/` first.

6 Datasets

The `datasets/` folder includes credit-fraud CSV, instruction JSONL for LoRA, and sample images for vision labs. Run `python download_datasets.py -minimal` to refresh bundled files, or `python download_datasets.py` to prefetch MNIST and Hugging Face datasets. Lab 09 downloads SST-2 from Hugging Face on first run (requires internet and a loaded session environment).

7 Spark Lab (Module 18 Only)

Spark needs JDK 17 and, on Windows, `HADOOP_HOME` pointing at `tools/hadoop`. Session launchers set `JAVA_HOME` when a JDK is found. If Spark fails with a `winutils` or `UnsatisfiedLinkError` message, confirm `tools/hadoop/bin/winutils.exe` exists and restart the terminal after sourcing the session script.

Table 1: Module, study PDF, and lab notebook.

Mod.	PDF	Notebook
00	00-environment-setup.pdf	(this guide)
01	01-rnns-sequence-modeling.pdf	lab01_rnn_sequence_modeling.ipynb
02	02-lstm-gru.pdf	lab02_lstm_gru_comparison.ipynb
03	03-attention-seq2seq.pdf	lab03_attention_seq2seq.ipynb
04	04-transformers.pdf	lab04_transformer_block.ipynb
05	05-gan-fundamentals.pdf	lab05_gan_mnist.ipynb
06	06-dcgan-gan-training.pdf	lab06_dcgan_mnist.ipynb
07	07-diffusion-theory.pdf	lab07_diffusion_visualization.ipynb
08	08-ddpm-stable-diffusion.pdf	lab08_ddpm_mnist.ipynb
09	09-bert.pdf	lab09_bert_finetune.ipynb
10	10-gpt-causal-lms.pdf	lab10_gpt_generation.ipynb
11	11-lora-peft.pdf	lab11_lora_finetune.ipynb
12	12-faster-rcnn.pdf	lab12_faster_rcnn_inference.ipynb
13	13-yolo-map.pdf	lab13_yolo_training.ipynb
14	14-segmentation-unet-maskrcnn.pdf	lab14_segmentation.ipynb
15	15-rl-mdps-qlearning.pdf	lab15_qlearning_frozenlake.ipynb
16	16-dqn.pdf	lab16_dqn_cartpole.ipynb
17	17-policy-gradients-actor-critic.pdf	lab17_a2c_cartpole.ipynb
18	18-spark-mllib.pdf	lab18_spark_mllib.ipynb
19	19-ddp-mlops.pdf	lab19_ddp_mlflow.ipynb
20	20-shap-lime-xai-deployment.pdf	lab20_xai_deployment.ipynb

8 MLflow (Lab 19)

`env.ps1` starts MLflow UI automatically at <http://127.0.0.1:5050> (SQLite store under `mlruns/mlflow.db`). If you used `-EnvOnly`, start it manually:

```
python -m mlflow ui --backend-store-uri sqlite:///mlruns/mlflow.db --port 5050
```

9 Troubleshooting

9.1 CUDA out of memory

Lower `batch_size` in the notebook. Modules 09 to 11 list safe values for 4 GB GPUs. Close browser GPU tabs and other CUDA applications.

9.2 Missing Python package

Activate `.venv`, then: `pip install -r requirements.txt`.

9.3 Wrong Jupyter kernel

Kernel menu → Change kernel → **Python 3.11 (labs)**. If missing, run:

```
python configure_kernel.py
```

9.4 Hugging Face download fails

Run `env.ps1` first. Check internet access. Lab 09 uses dataset `SetFit/sst2` (not GLUE SST-2).

9.5 Package list

`requirements.txt` includes PyTorch, transformers, diffusers, ultralytics, gymnasium, pyspark, mlflow, shap, lime, jupyter, and standard scientific Python libraries. Do not upgrade packages mid-course unless a lab explicitly requires it.

Source certification

Environment steps follow official documentation from Python.org, PyTorch.org, Hugging Face, Apache Spark, and MLflow. Hardware and disk estimates are conservative lab defaults for local Windows setups, not vendor SLAs. CUDA wheel selection matches PyTorch’s published CUDA 12.1 index at the time of writing. Spark Windows notes follow Apache Spark’s documented `winutils` requirement. MLflow tracking uses the SQLite backend documented in MLflow 3.x (filesystem `./mlruns` alone is deprecated).